

Stabilization of Markovian Jump Boolean Control Networks via Event-Triggered Control

Bingquan Chen[✉], Jinde Cao[✉], *Fellow, IEEE*, Guoping Lu[✉], and Leszek Rutkowski[✉], *Fellow, IEEE*

Abstract—In this article, we investigate the stabilization of Markovian jump Boolean control networks by a kind of event-triggered control, which is essentially an intermittent control scheme. First, a novel condition for the stability of Markovian jump Boolean networks is obtained based on the recurrence of finite-state homogeneous Markov chains. After that, the necessary and sufficient conditions for the stabilization of a Markovian jump Boolean control network by an event-triggered control are proposed based on an associated digraph, and the design method of the corresponding event-triggered control is given. Furthermore, in order to save control costs, we construct an event-triggered control with a minimal event-triggering set for the stabilization of the Markovian jump Boolean control network by finding a minimum-weight spanning branching forest of the associated digraph. Finally, an example is given to illustrate the effectiveness of the obtained results.

Index Terms—Event-triggered control (ETC), Markovian jump Boolean control networks (MJBCNs), semi-tensor product, stability, stabilization.

I. INTRODUCTION

Boolean networks (BNs) are discrete-time logical systems proposed by Kauffman [1] to model gene regulatory networks (GRNs). In a BN, there are only two possibilities for the current state of each node: ON (1) and OFF (0), which is determined by the previous states of neighbor nodes through logical relationships. Moreover, Boolean control networks (BCNs) are obtained by adding control inputs to

Manuscript received 14 April 2021; revised 21 September 2021; accepted 3 February 2022. Date of publication 14 February 2022; date of current version 30 January 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61573096 and Grant 61833005, in part by the Jiangsu Provincial Key Laboratory of Networked Collective Intelligence under Grant BM2017002, in part by the Scientific Research Foundation of Graduate School of Southeast University under Grant YBPY2136, and in part by the China Scholarships Council under Grant 202006090186. Recommended by Associate Editor Q. S. Jia. (*Corresponding author: Jinde Cao.*)

Bingquan Chen is with the Jiangsu Provincial Key Laboratory of Networked Collective Intelligence, School of Mathematics, Southeast University, Nanjing 210096, China (e-mail: bqchen1995@hotmail.com).

Jinde Cao is with the School of Mathematics and the Frontiers Science Center for Mobile Information Communication and Security, Southeast University, Nanjing 210096, China, with the Purple Mountain Laboratories, Nanjing 211111, China, and also with the Yonsei Frontier Lab, Yonsei University, Seoul 03722, South Korea (e-mail: jd-caoc@seu.edu.cn).

Guoping Lu is with the Institute of Systems Science, School of Electrical Engineering, Nantong University, Nantong 226019, China (e-mail: lu.gp@ntu.edu.cn).

Leszek Rutkowski is with the Systems Research Institute of the Polish Academy of Sciences, 01-447 Warsaw, Poland, and also with the Institute of Computer Science, AGH University of Science and Technology, 30-059 Kraków, Poland (e-mail: leszek.rutkowski@ibspan.waw.pl).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TAC.2022.3151237>.

Digital Object Identifier 10.1109/TAC.2022.3151237

BNs. Recently, BNs and BCNs have received extensive attentions from scholars. Especially after the semi-tensor product (STP) was invented by Cheng *et al.* [2], many control problems of BNs and BCNs have been investigated, such as stability and stabilization [3]–[14], controllability [15], [16], output tracking control [17], [18], and robust control problem [19], [20].

Gene regulation process is a noisy process in cells, which are often subject to significant internal fluctuations and external interference [21]. Consequently, it is better to view gene expression as a random process. Markov chain has been widely used as a general framework for modeling GRNs. In [22], a finite-state homogeneous Markov chain was constructed from microarray data, and it was shown that the Markov chain simulation emulated well the dynamic behavior of a small GRN. Probabilistic Boolean networks (PBNs) are the random models for GRNs proposed by Shmulevich *et al.* [23]. As shown in [12], PBNs can also be transformed into finite-state homogeneous Markov chains. If we view the dynamic behavior of a small subnetwork of a GRN as a Markov process, then the rest can be modeled as a Markovian switching system.

As we know, stability and stabilization are important issues because we always want to know whether a system can reach an ideal state. The stability of Markovian jump GRNs with time delays has been investigated in [24] and [25], where the GRNs were modeled by differential equations. Since BNs are also important models of GRNs, it is meaningful to study the stability and stabilization of Markovian jump Boolean networks (MJBNs). The stochastic stability of MJBNs has been studied in [9]. Furthermore, the finite-time stability of MJBNs has been investigated in [10] and [11]. Moreover, Guo *et al.* [12] studied the stability of MJBNs by converting MJBNs into finite-state homogeneous Markov chains. In addition, the stabilization of Markovian jump Boolean control networks (MJBCNs) by state feedback control and output state feedback control have been studied in [13] and [14], respectively.

In the study of GRNs, the traditional feedback control that is triggered at all the times may lead to some adverse results, such as the too frequent transmission of information between mRNA and protein [26], [27]. Thus, it is necessary to consider a control mechanism based on the measurable states of partial discrete times. As described in [28], an event-triggered control (ETC) is activated only under certain conditions, and it is not required to perform sensing and actuation periodically like a sampled-data control. Because of these advantages, the ETC technology has been widely used in many fields, such as spacecraft systems [29] and evolutionary networked games [30]. In addition, the event-based optimal control of discrete event systems has been studied and applied to air-conditioning systems [31]–[33]. Recently, ETCs have also been applied to BCNs and logical control networks (LCNs). In [4], a feasible ETC with a minimal event-triggering set was constructed for the stabilization of an LCN based on Edmonds's algorithm [34]. In addition, the finite-time stabilization of PBCNs by ETCs was researched in [6]. The robust ETC of BCNs has also been investigated in [20].

TABLE I
NOTATIONS

Notations	Definitions
$ S $	Cardinality of set S
$\mathbb{R}, \mathbb{N}, \mathbb{N}_+$	Set of real numbers, natural numbers, positive integers
$[n : m] (n < m)$	Set of integers $\{n, n+1, \dots, m\}$
$\mathbb{R}_{n \times m}$	Set of $n \times m$ real matrices
$\mathbb{R}^n$	Set of n -dimensional real vectors
$[L]_{ij}$	(i, j) -th entry of matrix L
$\text{Row}_i(L)$	i -th row of matrix L
$\text{Col}_i(L)$	i -th column of matrix L
$[v]_i$	i -th component of vector v
$v > 0$	Short for $[v]_i > 0 \forall i$
δ_n^i	i -th column of $n \times n$ identity matrix I_n
Δ_n	Set of columns of I_n
$\mathcal{L}_{n \times m}$	Set of $n \times m$ logical matrices
$\delta_n[i_1 \ i_2 \ \dots \ i_m]$	Simplification for logical matrix $[\delta_n^{i_1} \ \delta_n^{i_2} \ \dots \ \delta_n^{i_m}]$
$\mathbf{1}_n$	$\underbrace{[1 \ 1 \ \dots \ 1]^T}_n$

Motivated by the previous discussions, we study the stability of MJBNs and the stabilization of MJBCNs via ETCs. Unlike that in [9], the Markov chain that consists of the sequence of switching signals in this article may be nonergodic. Furthermore, unlike what was assumed in [11], the preferred state is not required to be a fixed point for all the modes. In addition, different from [12], we do not need to calculate the largest invariant subset when we verify the stability of an MJBN, and the obtained stability condition is more helpful than those in [12] for us to study the stabilization of MJBCNs via ETCs. The ETC in this article is derived from a kind of state-triggered intermittent control, which is proposed in [30] and improved in [4]. The ETCs in [4] and [30] are more suitable for time-invariant systems. Therefore, we extend this kind of ETC to a mode-dependent form and apply it to MJBCNs. The main contributions of this article are highlighted as follows.

- 1) A novel result on the stability of an MJBN is presented, which is helpful for us to study the stabilization of an MJBCN by an ETC.
- 2) The necessary and sufficient conditions for the stabilization of an MJBCN by an ETC are obtained.
- 3) A feasible ETC with a minimal event-triggering set for the stabilization of an MJBCN is constructed.

The rest of this article is organized as follows. Section II introduces some preliminaries. Section III contains the main results. In order to better illustrate the obtained results, an example is presented in Section IV. Finally, Section V concludes this article.

II. PRELIMINARIES

A. Notations

The basic notations to be used in the sequel are given in Table I. In addition, an $n \times m$ matrix F is called a stochastic matrix if $\mathbf{1}_n^T F = \mathbf{1}_m^T$.

B. Semi-tensor Product and Khatri-Rao Product

Definition 1 ([2]): Given $F \in \mathbb{R}_{n \times m}$ and $L \in \mathbb{R}_{p \times q}$, the STP of F and L , denoted by $F \ltimes L$, is defined as

$$F \ltimes L = (F \otimes I_m)(L \otimes I_p) \quad (1)$$

where $\otimes$ is the Kronecker product, and α represent the least common multiple of m and p . When $m = p$, the STP of F and L degenerates to the conventional product of F and L . Thus, the symbol $\ltimes$ can be omitted for simplicity when there is no ambiguity.

Proposition 1 ([2]):

- 1) For any $L \in \mathbb{R}_{n \times m}$ and $X \in \mathbb{R}^p$, $X \ltimes L = (I_p \otimes L) \ltimes X$.

- 2) For any $X \in \Delta_n$, $X \ltimes X = \Phi_n X$, where Φ_n is called the power reducing matrix.
- 3) For any $X \in \Delta_n$ and $Y \in \Delta_m$, $XY = W_{[m,n]} Y X$, where $W_{[m,n]}$ is called the swap matrix.

Definition 2 ([35]): Given $F \in \mathbb{R}_{n \times m}$ and $L \in \mathbb{R}_{p \times m}$, the Khatri-Rao product of F and L , denoted by $F * L$, is defined as by column as follows:

$$\text{Col}_i(F * L) = \text{Col}_i(F) \ltimes \text{Col}_i(L) \quad \forall i \in [1 : m].$$

C. Finite-State Homogeneous Markov Chains

A finite-state homogeneous Markov chain is a sequence of states ω_α , $\alpha \in \mathbb{N}$ with a state space $\mathcal{W} = \{s_1, s_2, \dots, s_{|\mathcal{W}|}\}$ and a transition probability matrix $\mathbb{P} = (p_{ij})$, i.e., $\Pr\{\omega_{\alpha+1} = s_j \mid \omega_\alpha = s_i\} = p_{ij} = [\mathbb{P}]_{ij} \forall \alpha \in \mathbb{N}$; moreover, $p_{ij}^{(k)} := \Pr\{\omega_{\alpha+k} = s_j \mid \omega_\alpha = s_i\}$ is termed as the k -step transition probability of ω_α , $\alpha \in \mathbb{N}$. A subset $S \subseteq \mathcal{W}$ is called an *invariant subset* of $\mathcal{W}$ with respect to ω_α if for any $s_i \in S$, it holds that $\Pr\{\omega_{\alpha+1} \in S \mid \omega_\alpha = s_i\} = 1$. Given $S \subseteq \mathcal{W}$, denote by $I_\omega(S)$ the *largest invariant subset* of $\mathcal{W}$ with respect to ω_α that is contained in S . It is easy to find that S is an invariant subset of $\mathcal{W}$ if and only if $I_\omega(S) = S$.

s_j is said to be *1-step reachable* from s_i , denoted by $s_i \rightarrow s_j$, if $p_{ij} > 0$. A sequence of states with arrows $s_0 \rightarrow s_1 \rightarrow s_2 \rightarrow \dots \rightarrow s_k$ is called an *admissible path* of length k from s_0 to s_k . Moreover, s_j is said to be *reachable* from s_i if there is an admissible path from s_i to s_j . Furthermore, two states s_i and s_j are said to be *connected* if they are reachable from each other. A subset $S \subseteq \mathcal{W}$ is called a *communicating class* of ω_α , $\alpha \in \mathbb{N}$, if every state in S is connected with any other in S and is not connected with any one that is not in S .

Given $s_i \in \mathcal{W}$, if set $\{k \geq 1 \mid p_{ii}^{(k)} > 0\}$ is nonempty, denote by $\phi(i)$ the greatest common divisor of all the elements in this set. Define the *hitting probabilities* of ω_α , $\alpha \in \mathbb{N}$, as $q_{ij}^{(k)} = \Pr\{\omega_k = s_j, \omega_\tau \neq s_j \forall \tau \in [1 : k-1] \mid \omega_0 = s_i\}$, where $k \geq 1$ and $s_i, s_j \in \mathcal{W}$. Let $\mathbf{q}_{ij} = \sum_{k=1}^{\infty} q_{ij}^{(k)}$. A state $s_i \in \mathcal{W}$ is called a *recurrent state* if $\mathbf{q}_{ii} = 1$, and is called a *transient state* if $\mathbf{q}_{ii} < 1$. If s_i is a recurrent state, let $\nu_i = \sum_{k=1}^{\infty} k q_{ii}^{(k)}$, and then s_i is called a *positive recurrent state* if $\nu_i < \infty$, and is called a *null recurrent state* if $\nu_i = \infty$.

Proposition 2 ([36]): Consider the Markov chain ω_α , $\alpha \in \mathbb{N}$ introduced previously. The set of transient states and the set of recurrent states are denoted by $\mathcal{B}$ and $\mathcal{C}$, respectively. Then the following statements hold.

- 1) There is no null recurrent state for a finite-state homogeneous Markov chain, so $\nu_i < \infty \forall s_i \in \mathcal{C}$.
- 2) If s_i is a recurrent state, $\lim_{n \rightarrow \infty} p_{ii}^{(n\phi)} = \frac{\phi}{\nu_i}$, where $\phi := \phi(i)$.
- 3) $\mathcal{W}$ can be decomposed into disjoint subsets as $\mathcal{W} = \mathcal{B} \cup \mathcal{C} = \mathcal{B} \cup \mathcal{C}_1 \cup \dots \cup \mathcal{C}_\ell$, where each $\mathcal{C}_k$ is a communicating class and an invariant subset of $\mathcal{W}$, $k = 1, \dots, \ell$, and $\mathcal{C}$ is an invariant subset of $\mathcal{W}$ but $\mathcal{B}$ is not.

Next, we introduce the definition, and necessary and sufficient conditions for the stability of finite-state homogeneous Markov chain ω_α , $\alpha \in \mathbb{N}$.

Definition 3: Given a subset $S \subseteq \mathcal{W}$, homogeneous Markov chain ω_α , $\alpha \in \mathbb{N}$ is said to be *S-stable*, if for any $\varpi \in \mathcal{W}$, $\lim_{\alpha \rightarrow \infty} \Pr\{\omega_\alpha \in S \mid \omega_0 = \varpi\} = 1$.

Lemma 1 ([12]): Given $S \subseteq \mathcal{W}$, the following four statements are equivalent.

- 1) Markov chain ω_α , $\alpha \in \mathbb{N}$ is *S-stable*.
- 2) $I_\omega(S) \neq \emptyset$ and ω_α , $\alpha \in \mathbb{N}$ is $I_\omega(S)$ -stable.

3) There exists a $k \leq |\mathcal{W}| - |I_\omega(S)|$ such that

$$\sum_{s_i \in I_\omega(S)} \text{Row}_i \left((\mathbb{P}^\tau)^k \right) > 0. \quad (2)$$

4) For any $s_j \in \mathcal{W} \setminus I_\omega(S)$, there exists an admissible path from s_j to (a state in) $I_\omega(S)$ no longer than $|\mathcal{W}| - |I_\omega(S)|$.

D. Digraph Theories

A *digraph* D is an ordered 3-tuple $(V(D), A(D), \psi_D)$, which consists of a set $V(D)$ of *vertices*, a set $A(D)$ of *arcs*, and an *incidence function* ψ_D that associates each arc with an ordered pair of vertices. If $\psi_D(a) = (i, j)$ for some $a \in A(D)$, we say a *joins* i to j or that i *dominates* j through a . Such a is called an *in-arc* of j and a *out-arc* of i , and moreover, i and j are called the *tail* and the *head* of a , respectively. The *indegree* and *outdegree* of i are the number of in-arcs and out-arcs of i , respectively. A sequence of arcs $(i_0, i_1), (i_1, i_2), \dots, (i_{k-1}, i_k)$ is called a *walk* from i_0 to i_k . A *cycle* is a walk in which only the initial and final vertices coincide. D is called a *strongly connected digraph* if for any $i, j \in V(D)$, there is a walk from i to j and a walk from j to i .

D' is called a *subgraph* of D if $V(D') \subseteq V(D)$, $A(D') \subseteq A(D)$, and $\psi_{D'}(a) = \psi_D(a) \forall a \in A(D')$. Furthermore, if $V(D') = V(D)$, then D' is called a *spanning subgraph* of D . The number of arcs, whose tail and head belong to $V(D')$ and $V(D) \setminus V(D')$, respectively, is termed the *outdegree* of D' . In D , *shrinking* D' means identifying D' into a super vertex. A strongly connected component (SCC) of D is a maximal strongly connected subgraph of D .

A *tree* is a connected acyclic undirected graph. Given root r , an *r -branching* is an orientation of a tree in which r has indegree zero and other vertices have indegree one. A *spanning branching forest* of D is a spanning subgraph of D consists of disjoint branchings. Add a *weight function* w that associates each arc of D with a nonnegative real number. A *minimum-weight spanning branching forest* of D with the set $\mathcal{R} \in V(D)$ of roots is a spanning branching forest of D with the set $\mathcal{R}$ of roots and with the minimum total weight of arcs.

III. MAIN RESULTS

The main results include two parts. The stability condition of MJBNs obtained in Section III-A is an important basis for studying the stabilization of MJBCNs by the ETC in Section III-B.

A. Stability of an MJBN

Consider an MJBN with n nodes and z possible realizations in an algebraic form as follows:

$$x(t+1) = F_{\sigma(t)} x(t) \quad (3)$$

where $x(t) \in \Delta_{2^n}$ is the state of the MJBN, $\sigma(t) \in [1 : z]$ is a switching signal variable, and $F_k \in \mathcal{L}_{2^n \times 2^n}$, $k \in [1 : z]$. A state $x_e \in \Delta_{2^n}$ is called a *fixed point* of system (3) for k th mode if $F_k x_e = x_e$.

We suppose that the sequence of switching signals $\sigma(t)$, $t \in \mathbb{N}$, is a homogeneous Markov chain with a transition probability matrix P , which may be nonergodic. Moreover, denote by C the set of recurrent states of $\sigma(t)$, $t \in \mathbb{N}$.

The probability transition digraph of $\sigma(t)$, $t \in \mathbb{N}$, is denoted by D_σ . There is an arc $(i, j) \in A(D_\sigma)$ if and only if $[P]_{ij} > 0$. It is easy to find that each communicating class of $\sigma(t)$, $t \in \mathbb{N}$, corresponds to an SCC of D_σ . By Proposition 2, C is equal to the set of vertices that belong to the SCCs with outdegree zero of D_σ . If z is big, we can calculate C by Algorithm 1, which takes at most $O(z^2)$ (see the Appendix for details).

Define the vector form of $\sigma(t)$ as $\theta(t) = \delta_z^{\sigma(t)}$. Then $\theta(t)$, $t \in \mathbb{N}$, is a homogeneous Markov chain with state space Δ_z and transition probability matrix P . Define $\tilde{F} = [F_1 \ F_2 \ \dots \ F_z]$, which is stacked by $F_1, F_2, \dots, F_z$. Then system (3) is equivalent to

$$x(t+1) = \tilde{F}\theta(t)x(t). \quad (4)$$

Definition 4: Given $x_e \in \Delta_{2^n}$, MJBN (3) is said to be x_e -stable if for any $x_0 \in \Delta_{2^n}$ and $\sigma_0 \in [1 : z]$, it holds that $\lim_{t \rightarrow \infty} \Pr\{x(t) = x_e \mid x(0) = x_0, \sigma(0) = \sigma_0\} = 1$.

Let $N = z2^n$. An instrumental variable, denoted by $y(t)$, is defined as

$$y(t) = \theta(t) \ltimes x(t) \in \Delta_N.$$

It is easy to prove that for any $\delta_z^i \in \Delta_z$ and $\delta_N^j \in \Delta_N$

$$\Pr\{\theta(t+1) = \delta_z^i \mid y(t) = \delta_N^j\} = [P^\tau \otimes \mathbf{1}_{2^n}]_{ij}. \quad (5)$$

Define an $N \times N$ stochastic matrix as $\hat{F} = (P^\tau \otimes \mathbf{1}_{2^n}) * \tilde{F}$. By (4) and (5), for any $\delta_N^i, \delta_N^j \in \Delta_N$

$$\Pr\{y(t+1) = \delta_N^i \mid y(t) = \delta_N^j\} = [\hat{F}]_{ij} \quad (6)$$

i.e., $y(t)$, $t \in \mathbb{N}$, is a homogeneous Markov chain with transition probability matrix $\hat{F}^\tau$. Therefore, it is much easier to study the stability of $y(t)$, $t \in \mathbb{N}$, than to study the stability of MJBN (3) directly.

Remark 1: Define $\mathcal{M}_e = \{\delta_z^i \ltimes x_e \mid i \in [1 : z]\}$. It is easy to prove that MJBN (3) is x_e -stable if and only if $y(t)$, $t \in \mathbb{N}$, is $\mathcal{M}_e$ -stable. Denote by $I_y(\mathcal{M}_e)$ the largest invariant subset of Δ_N with respect to $y(t)$ that is contained in $\mathcal{M}_e$. We can judge whether MJBN (3) is x_e -stable by inequality (2) after we obtain $I_y(\mathcal{M}_e)$, which can be calculated by the algorithm in [12], whose computational complexity is $O(|\mathcal{M}_e|^3)$ ($= O(z^3)$). This is greater than $O(z^2)$ required by calculating set C .

Next, a theorem is given to verify the stability of MJBN (3) without getting $I_y(\mathcal{M}_e)$. The first condition of the following theorem means that x_e is a fixed point of MJBN (3) for the modes whose indices belong to C .

Theorem 1: Given $x_e \in \Delta_{2^n}$, let $\mathcal{M}_e^c = \{\delta_z^i \ltimes x_e \mid i \in C\}$. Then MJBN (3) is x_e -stable if and only if

- 1) $F_i x_e = \tilde{F} \delta_z^i x_e = x_e \forall i \in C$;
- 2) there exists a $k \leq N - |\mathcal{M}_e^c|$ such that

$$\sum_{\delta_N^i \in \mathcal{M}_e^c} \text{Row}_i \left(\hat{F}^k \right) > 0$$

i.e., for any $y_0 \in \Delta_N \setminus \mathcal{M}_e^c$, there exists an admissible path from y_0 to $\mathcal{M}_e^c$ no longer than $N - |\mathcal{M}_e^c|$.

Proof: [Sufficiency] Suppose that the two conditions hold. Based on Lemma 1, we only need to prove that $\mathcal{M}_e^c$ is an invariant subset of Δ_N with respect to $y(t)$. According to Proposition 2, C is an invariant subset of $[1 : z]$ with respect to $\sigma(t)$. Therefore, for any $\delta_z^i \ltimes x_e \in \mathcal{M}_e^c$, it holds that $\Pr\{y(t+1) \in \mathcal{M}_e^c \mid y(t) = \delta_z^i \ltimes x_e\} = \Pr\{x(t+1) = x_e, \sigma(t+1) \in C \mid x(t) = x_e, \sigma(t) = i\} = \Pr\{F_i x_e = x_e\} = 1$. Thus, $\mathcal{M}_e^c$ is an invariant subset of Δ_N , i.e., $I_y(\mathcal{M}_e^c) = \mathcal{M}_e^c$. Then by Lemma 1, $y(t)$, $t \in \mathbb{N}$, is $\mathcal{M}_e^c$ -stable, which implies that MJBN (3) is x_e -stable.

[Necessity] Suppose that MJBN (3) is x_e -stable. For any $i \in C$, $x_0 \in \Delta_{2^n}$ and $k \in \mathbb{N}_+$

$$\begin{aligned} & \Pr\{x(k\phi + 1) = x_e \mid x(0) = x_0, \sigma(0) = i\} \\ & \leq \Pr\{y(k\phi) = \delta_z^i \times x_e \mid x(0) = x_0, \sigma(0) = i\} \\ & \times \Pr\{x(k\phi + 1) = x_e \mid y(k\phi) = \delta_z^i \times x_e\} \\ & + \sum_{j \neq i} \Pr\{y(k\phi) = \delta_z^j \times x_e \mid x(0) = x_0, \sigma(0) = i\} \\ & \times \Pr\{x(k\phi + 1) \neq x_e \mid y(k\phi) = \delta_z^j \times x_e\} \\ & \leq \Pr\{\sigma(k\phi) = i \mid \sigma(0) = i\} \times \Pr\{F_i x_e = x_e\} \\ & + \Pr\{\sigma(k\phi) \neq i \mid \sigma(0) = i\} \\ & + \Pr\{x(k\phi) \neq x_e \mid x(0) = x_0, \sigma(0) = i\} \end{aligned}$$

where $\phi := \phi(i)$. Letting $k \rightarrow \infty$, by Proposition 2, $1 \leq \frac{\phi}{\nu_i} \times \Pr\{F_i x_e = x_e\} + (1 - \frac{\phi}{\nu_i}) + 0$, which implies that $F_i x_e = x_e$. By the arbitrariness of $i \in C$, the condition 1) holds. Next, similar to the proof of the sufficiency, we can prove that $I_y(\mathcal{M}_e^c) = \mathcal{M}_e^c$. Then by Lemma 1, the condition 2) holds. ■

B. Stabilization of an MJBCN via an ETC

In this article, the control mechanism of the ETC is that the system remains in MJBN (3) when the control is not triggered, and only in some preset cases, the system is controlled by m external inputs as follows:

$$x(t+1) = L_{\sigma(t)} u(t) x(t) \quad (7)$$

where $u(t) \in \Delta_{2^m}$ is the control input, and $L_k \in \mathcal{L}_{2^n \times 2^n + m}$, $k \in [1 : z]$.

We suppose that $x(t)$ and $\sigma(t)$ are measurable. Then $y(t)$ is also measurable. The event-triggering set, denoted by $\Omega \subseteq \Delta_N$, consists of all the values of $y(t)$ that trigger the control. Combining MJBN (3) with MJBCN (7), a system under the ETC is obtained as

$$x(t+1) = \begin{cases} F_{\sigma(t)} x(t), & \text{if } y(t) \in \Delta_N \setminus \Omega \\ L_{\sigma(t)} u(t) x(t), & \text{if } y(t) \in \Omega. \end{cases} \quad (8)$$

Moreover, $u(t)$ is assumed to follow a feedback mechanism

$$u(t) = K y(t) \quad (9)$$

where $K \in \mathcal{L}_{2^m \times N}$. The ETC is a kind of intermittent control, which is uniquely determined by a feedback mechanism (9) and event-triggering set Ω . Compared with the traditional state feedback control, the ETC can reduce the time of control and save the control costs.

Next, we give an application scenario of this kind of ETC in biology. In a GRN, we view the cluster of gene nodes we are concerned as a subnetwork. The dynamic behavior of the rest can be regarded as a Markov process, denoted by $\sigma(t)$, $t \in \mathbb{N}$. Then, the subnetwork can be modeled as MJBN (3), where $x(t)$ denotes the state of the concerned genes. Suppose that the states of all the genes are measurable. When the genes are in certain states, we add related drugs for the concerned genes. In this case, the subnetwork can be modeled as MJBCN (7), where $u(t)$ represents the dose of drugs.

Let $M = 2^m + 1$. In order to get a unified expression of (8), an M -dimensional vector $\tilde{u}(t)$ is defined as

$$\tilde{u}(t) = \begin{cases} \delta_M^M, & \text{if } y(t) \in \Delta_N \setminus \Omega \\ [u(t)^\top \ 0]^\top, & \text{if } y(t) \in \Omega. \end{cases} \quad (10)$$

It is easy to find that $\tilde{u}(t) = \delta_M^j$ if and only if $u(t) = \delta_{2^m}^j$ when $j \leq 2^m$, and $\tilde{u}(t) = \delta_M^M$ means that there is “no control” at time t . Define $G_i = [L_i \ F_i]$ stacked by L_i and F_i , $i \in [1 : z]$. Then system (8) is equivalent to

$$x(t+1) = G_{\sigma(t)} \tilde{u}(t) x(t). \quad (11)$$

Define an $M \times N$ logical matrix $\tilde{K}$ by column as follows:

$$\text{Col}_i(\tilde{K}) = \begin{cases} \delta_M^M, & \text{if } \delta_N^i \in \Delta_N \setminus \Omega \\ [(\text{Col}_i(K))^\top \ 0]^\top, & \text{if } \delta_N^i \in \Omega. \end{cases} \quad (12)$$

Then the ETC of system (8) can be expressed as

$$\tilde{u}(t) = \tilde{K} y(t). \quad (13)$$

Conversely, if $\tilde{K}$ is given, we can obtain Ω as

$$\Omega = \left\{ \delta_N^i \mid \text{Col}_i(\tilde{K}) \neq \delta_M^M \right\} \quad (14)$$

and calculate K by column as follows:

$$\text{Col}_i(K) = \begin{cases} \delta_{2^m}^{\diamond}, & \text{if } \text{Col}_i(\tilde{K}) = \delta_M^M \\ \delta_{2^m}^k, & \text{if } \text{Col}_i(\tilde{K}) = \delta_M^k, k \leq 2^m \end{cases} \quad (15)$$

where $\diamond$ can be any one in $[1 : 2^m]$, which makes no difference because feedback control (9) is activated only when $y(t) \in \Omega$.

Define $\tilde{G} = [G_1 \ G_2 \ \dots \ G_z]$ stacked by $G_1, G_2, \dots, G_z$. Define a $2^n \times N$ logical matrix as $\tilde{H} = \tilde{G} W_{[M,z]} \tilde{K} \Phi_N$. By (11) and Proposition 1, direct calculations yield that

$$\begin{aligned} x(t+1) &= \tilde{G} \theta(t) \left(\tilde{K} \theta(t) x(t) \right) x(t) \\ &= \tilde{G} W_{[M,z]} \left(\tilde{K} \theta(t) x(t) \right) \theta(t) x(t) \\ &= \tilde{G} W_{[M,z]} \tilde{K} \Phi_N \theta(t) x(t) \\ &= \tilde{H} \theta(t) x(t). \end{aligned} \quad (16)$$

Thus, a closed-loop system is obtained from system (11) under ETC (13) as follows:

$$x(t+1) = \tilde{H} \theta(t) x(t). \quad (17)$$

Assume that the feedback matrix of ETC (13) is given as

$$\tilde{K} = \delta_M [\mu_1 \ \mu_2 \ \dots \ \mu_N]$$

where $\mu_1, \mu_2, \dots, \mu_N \in [1 : M]$. Split each G_i into M blocks as $G_i = [G_{i,1} \ G_{i,2} \ \dots \ G_{i,M}]$. Define $\lambda_{i,j} = \mu_{(i-1)2^n + j}$. Then according to (16), for any $\delta_z^i \in \Delta_z$ and $\delta_{2^n}^j \in \Delta_{2^n}$

$$\begin{aligned} \tilde{H} \delta_z^i \delta_{2^n}^j &= \tilde{G} \delta_z^i \left(\tilde{K} \delta_z^i \delta_{2^n}^j \right) \delta_{2^n}^j \\ &= \text{Col}_j \left(G_{i, \lambda_{i,j}} \right). \end{aligned} \quad (18)$$

Define an $N \times NM$ stochastic matrix H as

$$H = (\mathbf{1}_M^\top \otimes P^\top \otimes \mathbf{1}_{2^n}^\top) * \left(\tilde{G} W_{[M,z]} \right). \quad (19)$$

Split H into M blocks as $H = [H_1 \ H_2 \ \dots \ H_M]$. Similar to (6), it is easy to prove that for any $\delta_N^i, \delta_N^j \in \Delta_N$ and $\delta_M^k \in \Delta_M$

$$\Pr\{y(t+1) = \delta_N^i \mid y(t) = \delta_N^j, u(t) = \delta_M^k\} = [H_k]_{ij}. \quad (20)$$

Moreover, define an $N \times N$ stochastic matrix $\hat{H}$ as follows:

$$\text{Col}_i(\hat{H}) = \text{Col}_i(H_{\mu_i}), i \in [1 : N]. \quad (21)$$

Then by (20), under ETC (13), for any $\delta_N^i, \delta_N^j \in \Delta_N$

$$\Pr\{y(t+1) = \delta_N^i \mid y(t) = \delta_N^j\} = [H_{\mu_j}]_{ij} = [\hat{H}]_{ij} \quad (22)$$

i.e., $y(t)$, $t \in \mathbb{N}$, is a homogeneous Markov chain with transition probability matrix $\hat{H}^\top$.

Now, construct a digraph D_H associated with matrix H to describe the probability transitions of $y(t)$, $t \in \mathbb{N}$, under control. Let

$V(D_H) := [1 : N]$. Two functions $\mathcal{U}$ and w associate each arc of D_H with a control input and a nonnegative real number, respectively. For any $\delta_N^i, \delta_N^j \in \Delta_N$ ($i \neq j$), and $\delta_M^\mu \in \Delta_M$, there exists an arc $a \in A(D_H)$ such that $\psi_{D_H}(a) = (i, j)$ and $\mathcal{U}(a) = \delta_M^\mu$, if and only if $[H_\mu]_{ij} > 0$. Moreover, let $w(a) = 0$ if $\mu = M$ and $w(a) = 1$ if $\mu \leq M - 1$. If there are parallel arcs, we keep only one with the minimum weight.

Based on digraph D_H and Theorem 1, the following theorem provides the necessary conditions for the stabilization of system (11) via ETC (13).

Theorem 2: Given $x_e = \delta_{2^n}^d \in \Delta_{2^n}$, define $\mathcal{M}_e^c = \{\delta_z^i \times x_e \mid i \in C\}$ and $\mathcal{R} = \{j \mid \delta_N^j \in \mathcal{M}_e^c\}$. If system (11) is stabilized to x_e by ETC (13), the following two conditions hold.

- 1) For any given $i \in C$, there exists an $\eta_i \in [1 : M]$ such that $\text{Col}_d(G_{i, \eta_i}) = x_e$.
- 2) D_H has a spanning branching forest with the set $\mathcal{R}$ of roots.

Proof: Assume that system (11) is stabilized to x_e by ETC (13), i.e., closed-loop system (17) is x_e -stable. According to (18) and Theorem 1, $\text{Col}_d(G_{i, \eta_i}) = \hat{H} \delta_z^i \delta_{2^n}^d = \delta_{2^n}^d \forall i \in C$, where $\eta_i := \lambda_{i, d}$. It follows from (22) that $y(t)$, $t \in \mathbb{N}$, is a homogeneous Markov chain with transition probability matrix $\hat{H}^\top$. Given two different states δ_N^i and δ_N^j , if $\delta_N^j \rightarrow \delta_N^i$, then we have $[H_{\mu_j}]_{ij} = [\hat{H}]_{ij} > 0$, which implies that there exists an arc a in an associated digraph D_H that joins i to j and satisfies $\mathcal{U}(a) = \delta_M^{\mu_j}$. According to Theorem 1, for any $j \in [1 : N] \setminus \mathcal{R}$, there exists an admissible path $\delta_N^j \rightarrow \delta_N^{i_{k-1}} \rightarrow \dots \rightarrow \delta_N^{i_0}$ of $y(t)$, $t \in \mathbb{N}$, from state δ_N^j to state $\delta_N^{i_0} \in \mathcal{M}_e^c$, which corresponds to a walk $(i_0, i_1), (i_1, i_2), \dots, (i_{k-1}, j)$ from a root $i_0 \in \mathcal{R}$ to j in the associated digraph D_H . Thus, D_H has a spanning branching forest with the set $\mathcal{R}$ of roots. ■

An ETC in the form of (13) is said to be feasible for the stabilization of system (11) if it can stabilize system (11) to the target state. Next, we show that the sufficiency of Theorem 2 also holds and constructs a feasible ETC for the stabilization of system (11) under the conditions of Theorem 2.

Theorem 3: Suppose that the conditions 1) and 2) of Theorem 2 hold. Denote by $T_{\mathcal{R}}$ one of the spanning branching forest of D_H with the set $\mathcal{R}$ of roots. For any $j \in [1 : N] \setminus \mathcal{R}$, denote by a_j the unique in-arc of j in $T_{\mathcal{R}}$. Then system (11) is stabilized to $x_e = \delta_{2^n}^d$ by ETC (13) with

$$\tilde{K} = \delta_M[\mu_1 \mu_2 \dots \mu_N] \quad (23)$$

where

$$\begin{cases} \text{Col}_d(G_{i, \lambda_{i, d}}) = \delta_{2^n}^d, & \text{if } i \in C \\ \delta_M^{\mu_j} = \mathcal{U}(a_j), & \text{if } j \in [1 : N] \setminus \mathcal{R}. \end{cases} \quad (24)$$

Proof: It is easy to prove that $(i - 1)2^n + d \in \mathcal{R}$ if and only if $i \in C$, so all the columns of matrix (23) are obtained by (24). Let us consider a closed-loop system (17). For any $i \in C$, by (18) and (24), we have $\hat{H} \delta_z^i \delta_{2^n}^d = \text{Col}_d(G_{i, \lambda_{i, d}}) = \delta_{2^n}^d$. For any $j \in [1 : N] \setminus \mathcal{R}$, there exists a unique vertex i in $T_{\mathcal{R}}$ that dominates j through a_j . Moreover, since $\delta_M^{\mu_j} = \mathcal{U}(a_j)$, we have $[\hat{H}]_{ij} = [H_{\mu_j}]_{ij} > 0$, which implies that $\delta_N^j \rightarrow \delta_N^i$. Therefore, we can prove by induction that for any $j \in [1 : N] \setminus \mathcal{R}$, there is a walk $(i_0, i_1), (i_1, i_2), \dots, (i_{k-1}, j)$ no longer than $N - |\mathcal{M}_e^c|$ from one of the roots $i_0 \in \mathcal{R}$ to j in $T_{\mathcal{R}}$, which corresponds to an admissible path $\delta_N^j \rightarrow \delta_N^{i_{k-1}} \rightarrow \dots \rightarrow \delta_N^{i_0}$ of Markov chain $y(t)$, $t \in \mathbb{N}$, from δ_N^j to $\delta_N^{i_0} \in \mathcal{M}_e^c$. By Theorem 1, the closed-loop system (17) is x_e -stable; equivalently, system (11) is stabilized to x_e by ETC (13) with a feedback matrix (23). ■

Remark 2: For a BCN (or an LCN), as shown in [3], the system can be stabilized to $x_e = \delta_{2^n}^d$ if and only if x_e can be fixed under control and the associated digraph, describing the state transitions of $x(t)$, $t \in \mathbb{N}$, has a spanning branching rooted at d . The stabilization

problem of PBCNs can be treated similarly. But $x(t)$, $t \in \mathbb{N}$, is no longer a Markov chain for an MJBCN. Thus, associated digraph D_H is given to describe the probability transitions of $y(t)$, $t \in \mathbb{N}$. Theorems 2 and 3 have provided the relation between the associated digraph and the stabilization of an MJBCN by an ETC.

In order to save control costs, it is better to find a feasible ETC with a minimal event-triggering set for the stabilization of system (11). According to (14), we just need to maximize the number of columns of $\tilde{K}$ that are equal to δ_M^M , on the premise that system (11) can be stabilized to target state x_e . Since $w(a) = 0$ when $\mathcal{U}(a) = \delta_M^M$ and $w(a) = 1$ when $\mathcal{U}(a) \neq \delta_M^M$, if we find a minimum-weight spanning branching forest of D_H , then in the obtained forest, there must be as many arcs as possible to satisfy $\mathcal{U}(a) = \delta_M^M$.

Based on Theorem 3 and the previous discussions, it is easy to obtain the following corollary, whose proof is omitted.

Corollary 1: Given a target state $x_e = \delta_{2^n}^d \in \Delta_{2^n}$, if the two conditions of Theorem 2 hold, we can construct a feasible ETC of the form (13), with a minimal event-triggering set for the stabilization of system (11), through the following two steps.

- 1) For every $i \in C$, if $\text{Col}_d(G_{i, M}) = x_e$, let $\delta_M^{\eta_i} = \delta_M^M$; otherwise, let $\delta_M^{\eta_i}$ be any $\delta_M^k \in \Delta_M$ such that $\text{Col}_d(G_{i, k}) = x_e$, where $\eta_i := \lambda_{i, d}$.
- 2) Construct a minimum-weight spanning branching forest of D_H with the set $\mathcal{R}$ of roots, to obtain $\delta_M^{j_j}$, $j \in [1 : N] \setminus \mathcal{R}$, by Algorithm 2 (see the Appendix for details).

Remark 3: According to Remark 2, the stabilization problem via ETC becomes more complicated when the system is an MJBCN instead of an LCN studied in [4]. The object described by the associated digraph should change from $x(t)$, $t \in \mathbb{N}$, to $y(t)$, $t \in \mathbb{N}$. And as shown in Theorem 3, we have to find a spanning branching forest with the set $\mathcal{R}$ of roots rather than a spanning branching with single root d . It is based on Theorem 1 that the set $\mathcal{R}$ of roots can be found. Furthermore, in order to realize the steps of Corollary 1, we need to modify Edmonds's Algorithm [34] so that it can find a minimum-weight spanning branching forest with the multiple roots of D_H .

IV. SIMULATIONS

Example 1: Consider an MJBN with three nodes and two possible realizations as follows [37]:

$$\begin{cases} a_1(t+1) = f_1^{\sigma(t)}(a_1(t), a_2(t), a_3(t)) \\ a_2(t+1) = f_2^{\sigma(t)}(a_1(t), a_2(t), a_3(t)) \\ a_3(t+1) = f_3^{\sigma(t)}(a_1(t), a_2(t), a_3(t)) \end{cases} \quad (25)$$

where $(f_1^1, f_2^1, f_3^1) = (a_1, a_2, a_3)$, $(f_1^2, f_2^2, f_3^2) = (a_1, \neg a_1 \wedge a_3, a_3)$, and $a_1, a_2, a_3 \in \{0, 1\}$ denote the concentration levels of inhibitor of apoptosis proteins, active caspase 3 (C3a), active caspase 8 (C8a), respectively, and $\sigma(t) = \{1, 2\}$ is a switching signal. Let $x(t) = \delta_2^{2-a_1(t)} \times \delta_2^{2-a_2(t)} \times \delta_2^{2-a_3(t)} \in \Delta_8$. Based on STP, system (25) can be expressed as

$$x(t) = F_{\sigma(t)} x(t) \quad (26)$$

where $F_1 = \delta_8[1\ 2\ 3\ 4\ 5\ 6\ 7\ 8]$ and $F_2 = \delta_8[3\ 4\ 3\ 4\ 5\ 8\ 5\ 8]$. An alternative MJBCN is given as follows:

$$\begin{cases} a_1(t+1) = \mathbb{L}_1^{\sigma(t)}(a_1(t), a_2(t), a_3(t), b(t)) \\ a_2(t+1) = \mathbb{L}_2^{\sigma(t)}(a_1(t), a_2(t), a_3(t), b(t)) \\ a_3(t+1) = \mathbb{L}_3^{\sigma(t)}(a_1(t), a_2(t), a_3(t), b(t)) \end{cases} \quad (27)$$

where $(\mathbb{L}_1^1, \mathbb{L}_2^1, \mathbb{L}_3^1) = (\neg a_2 \vee b, a_1, \neg a_2 \wedge b)$, $(\mathbb{L}_1^2, \mathbb{L}_2^2, \mathbb{L}_3^2) = (a_2 \wedge b, \neg a_1 \wedge a_3, a_2 \vee b)$, and $b \in \{0, 1\}$ denotes the concentration level of tumor necrosis factor. Let $u(t) = \delta_2^{2-b(t)} \in \Delta_2$. Similarly, system (27)

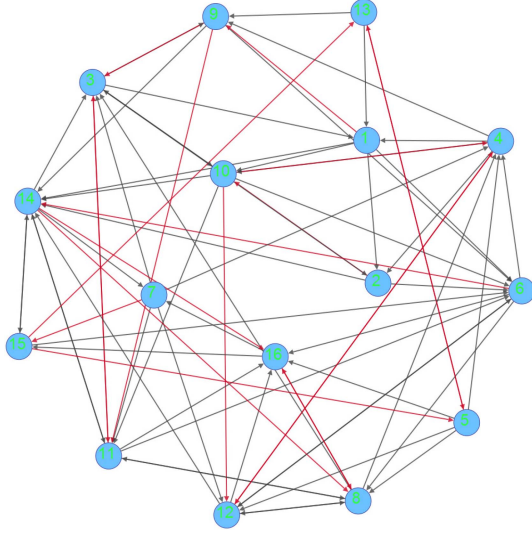


Fig. 1. Graphical representation of D_H , where red arrows denote the arcs with weight 0 and black arrows denote the arcs with weight 1.

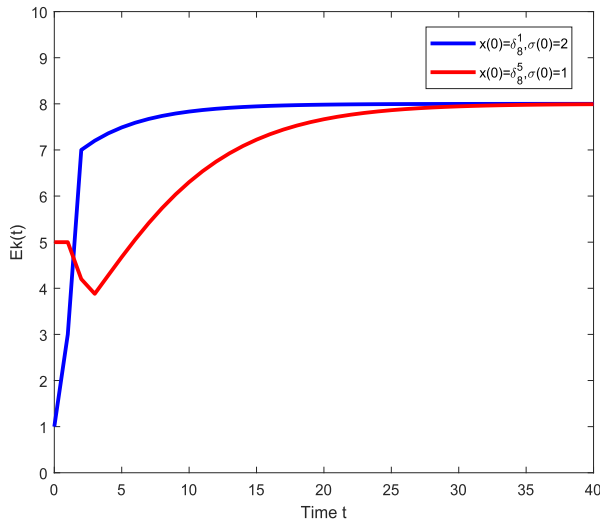


Fig. 2. Expectation of $k(t)$, where $x(t) = \delta_8^{k(t)}$.

is converted into

$$x(t) = L_{\sigma(t)} u(t) x(t) \quad (28)$$

where $L_1 = \delta_8[2211443366228844]$ and $L_2 = \delta_8[3377135777885768]$. It is easy to find that $N = 16$ and $M = 3$.

Assume that $\sigma(t)$, $t \in \mathbb{N}$, is a homogeneous Markov chain with transition probability matrix $P = \begin{pmatrix} 4/5 & 1/5 \\ 1/5 & 4/5 \end{pmatrix}$ and that the target state is $x_e = \delta_8^8$. It is easy to obtain that $C = \{1, 2\}$, which implies that $\mathcal{M}_e^c = \{\delta_{16}^8, \delta_{16}^{16}\}$. Let $G_i = [L_i F_i]$, $i = 1, 2$. Similar to (11), combining MJBN (26) with MJBCN (28), we can obtain a control system

$$x(t+1) = G_{\sigma(t)} \tilde{u}(t) x(t) \quad (29)$$

Algorithm 1: An Algorithm to Calculate C .

Input: D_σ , with sets $A_1, \dots, A_z$ and function h .
Output: The set C of recurrent states of $\sigma(t)$, $t \in \mathbb{N}$.

```

1 Initialize a stack  $\mathcal{S}$ ; num := cnt := 0;  $C = \emptyset$ ;
   out(i) := dfn(i) := low(i) := 0  $\forall i \in [1 : z]$ ;
2 for i := 1 to z do
3   if dfn(i) = 0 then
4     Tarjan(i);
5 for i := 1 to z do
6   for a  $\in A_i$  do
7     if id(i)  $\neq$  id(h(a)) then
8       out(id(i)) := out(id(i)) + 1;
9 for i := 1 to z do
10  if out(id(i)) = 0 then
11    C := C  $\cup$  {i};
12 return C;
13 Function Tarjan(i)
14   num := num + 1; dfn(i) := low(i) := num; vis(i) := 1;
15   Stack-In( $\mathcal{S}$ , i);
16   for a  $\in A_i$  do
17     j := h(a);
18     if dfn(j) = 0 then
19       Tarjan(j);
20     low(i) := min(low(i), low(j));
21   else if vis(j) = 1 then
22     low(i) := min(low(i), dfn(j));
23 if dfn(i) = low(i) then
24   cnt := cnt + 1;
25   repeat
26     j := Stack-Out( $\mathcal{S}$ );
27     vis(j) := 0; id(j) := cnt;
28   until i = j;

```

where $\tilde{u}(t) \in \Delta_3$ is defined as (10). By (19), we can get

$$\begin{aligned}
H = & \frac{4}{5} \delta_{16} [22114433111114149111315 \\
& 662288441414161613151416 \\
& 123456781112111213161316] \\
& + \frac{1}{5} \delta_{16} [1010991212111133661357 \\
& 141410101616121266885768 \\
& 91011121314151634345858].
\end{aligned}$$

It is easy to calculate that $\text{Col}_8(G_{1,3}) = \text{Col}_8(G_{2,3}) = \delta_8^8$. Construct digraph D_H as shown in Fig. 1. Let $\mathcal{R} = \{8, 16\}$. By Algorithm 2 and Corollary 1, we obtain an ETC

$$\tilde{u}(t) = \tilde{K} y(t) \quad (30)$$

with $\tilde{K} = \delta_3[333333313223333]$ that can stabilize system (29) to δ_8^8 . According to (14) and (15), we can get a minimal event-triggering set $\Omega = \{9, 11, 12\}$ and corresponding feedback matrix $K = \delta_2[\diamond \diamond \diamond \diamond \diamond \diamond \diamond \diamond 1 \diamond 22 \diamond \diamond \diamond \diamond]$, where each $\diamond$ can be either 1 or 2.

Let $\delta_8^{k(t)} = x(t)$. Since the state set is finite, for any $d \in [1 : 8]$, $\lim_{t \rightarrow +\infty} \mathbb{P}\{x(t) = \delta_8^d\} = 1$ if and only if $\lim_{t \rightarrow +\infty} \mathbb{E}k(t) = d$. Under ETC (30), the expectation of $k(t)$ at each time under initial conditions

Algorithm 2: An Algorithm to Help Design a Feasible ETC with a Minimal Event Triggering Set.

Input: D_H , together with functions $w, \mathcal{U}, h, e$.

Output: $\delta_M^{\mu j}$, $j \in [1 : N] \setminus \mathcal{R}$.

```

1 Initialize:  $\hat{h}(a) := h(a)$ ,  $\hat{e}(a) := e(a) \forall a \in A(D_H)$ ;  $\hat{N} := N$ ;
    $\hat{\mathcal{R}} := \mathcal{R}$ ; cnt := 1;
2 while cnt  $\neq$  0 do
3   cnt := 0;
4   for  $i := 1$  to  $\hat{N}$  do
5     in( $i$ ) := INF; in( $i$ ) := vis( $i$ ) := -1;
6   for  $a \in A(D_H)$  do
7     if  $w(a) < \text{in}(\hat{h}(a))$  and  $\hat{h}(a) \neq \hat{e}(a)$  then
8       pre( $\hat{h}(a)$ ) :=  $\hat{e}(a)$ ; in( $\hat{h}(a)$ ) :=  $w(a)$ ;
9        $\delta_M^{\mu h(a)} := \mathcal{U}(a)$ ;
10  if  $\{i \in [1 : \hat{N}] \setminus \hat{\mathcal{R}} \mid \text{in}(i) = \text{INF}\} \neq \emptyset$  then
11    return -1;
12  for  $i := 1$  to  $\hat{N}$  do
13     $v := i$ ;
14    while vis( $v$ )  $\neq$   $i$ , id( $v$ ) = -1 and  $v \notin \hat{\mathcal{R}}$  do
15      vis( $v$ ) :=  $i$ ;  $v := \text{pre}(v)$ ;
16    if  $v \notin \hat{\mathcal{R}}$  and id( $v$ ) = -1 then
17      cnt := cnt + 1;  $\hat{v} := v$ ; id( $v$ ) := cnt;
18      while pre( $\hat{v}$ )  $\neq$   $v$  do
19        id( $\hat{v}$ ) := cnt;  $\hat{v} := \text{pre}(\hat{v})$ ;
20  for  $i := 1$  to  $\hat{N}$  do
21    if id( $i$ ) = -1 then
22      cnt := cnt + 1; id( $i$ ) := cnt;
23  for  $a \in A(D_H)$  do
24     $v := \hat{h}(a)$ ;  $\hat{h}(a) := \text{id}(\hat{h}(a))$ ;  $\hat{e}(a) := \text{id}(\hat{e}(a))$ ;
25    if  $\hat{h}(a) \neq \hat{e}(a)$  then
26       $w(a) := w(a) - \text{in}(v)$ ;
27   $\hat{N} := \text{cnt}$ ;  $\hat{\mathcal{R}} := \{\text{id}(r) \mid r \in \hat{\mathcal{R}}\}$ ;
28 return  $\delta_M^{\mu j}$ ,  $j \in [1 : N] \setminus \mathcal{R}$ ;
```

$x(0) = \delta_8^1$, $\sigma(0) = 2$ and $x(0) = \delta_8^5$, $\sigma(0) = 1$, respectively, is shown in Fig. 2.

V. CONCLUSION

This article investigated the stabilization of MJBCNs by a class of ETC. After the recurrent states of the switching signal sequence were found, a novel condition was obtained for the stability of an MJBN. On the basis of this stability condition and the associated digraph, the necessary and sufficient criteria for the stabilization of an MJBCN by an ETC were proposed. Meanwhile, an ETC that can stabilize the system was obtained. Moreover, a feasible ETC with a minimal event-triggering set for the stabilization of the MJBCN was constructed by an algorithm, which was modified from Edmonds's algorithm. In future work, one can consider the synchronization and output tracking of MJBCNs by this kind of ETC.

APPENDIX ALGORITHMS

Algorithm 1, with time complexity $O(|V(D_\sigma)| + |A(D_\sigma)|) (\leq O(z^2))$, mainly includes two parts: 1) shrinking SCCs into super vertices by Tarjan Algorithm [38] (lines 2–4 and 13–28); 2) finding all the SCCs with outdegree zero.

In Algorithm 1, $h(a)$ represents the head of arc a ; $\text{out}(i)$ denotes the outdegree of the i th vertex (or super vertex); $\text{id}(i)$ denotes the SCC to which i belongs; $\text{Stack-In}(S, i)$ means pushing i to the top of S ; $j := \text{Stack-Out}(S)$ means assigning the top element of S to j and popping it from S ; A_i denotes the set of out-arcs of i in D_σ .

A minimum-weight spanning branching of a digraph can be found by Edmonds's Algorithm [34], which is not applicable when the system is an MJBCN instead of an LCN considered in [4]. Algorithm 2 is simply modified from Edmonds's Algorithm, with the same time complexity $O(|V(D_H)| \bullet |A(D_H)|)$, which can find a minimum-weight spanning branching forest of D_H . Each "while" loop mainly includes three steps: 1) constructing a spanning subgraph by selecting one arc with the minimum weight for every vertex (lines 6–8); 2) shrinking all the circles in the spanning subgraph into super vertices (line 12–22); 3) updating the weights of arcs (lines 23–26).

In Algorithm 2, $h(a)$ and $e(a)$ represent the head and tail of arc a , respectively; $\text{in}(i)$ denotes the weight of the in-arc of i in the spanning subgraph constructed by the step 1); $\text{pre}(i)$ represents the vertex dominating i in the spanning subgraph; $\text{id}(i)$ represents the index of the cycle to which i belongs; cnt counts the number of vertices after each round of shrinking; "return -1" means that there is no feasible ETC. The 9th line is the key operation to help design the ETC.

REFERENCES

- [1] S. A. Kauffman, "Metabolic stability and epigenesis in randomly constructed genetic nets," *J. Theor. Biol.*, vol. 22, no. 3, pp. 437–467, 1969.
- [2] D. Cheng, H. Qi, and Z. Li, *Analysis and Control of Boolean Networks: A Semi-Tensor Product Approach*. London, U.K.: Springer, 2011.
- [3] J. Liang, H. Chen, and Y. Liu, "On algorithms for state feedback stabilization of Boolean control networks," *Automatica*, vol. 84, pp. 10–16, 2017.
- [4] S. Zhu, Y. Liu, Y. Lou, and J. Cao, "Stabilization of logical control networks: An event-triggered control approach," *Sci. China Inf. Sci.*, vol. 63, Jan. 2020, Art. no. 112203.
- [5] B. Chen, J. Cao, G. Lu, and L. Rutkowski, "Lyapunov functions for the set stability and the synchronization of Boolean control networks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 67, no. 11, pp. 2537–2541, Nov. 2020.
- [6] S. Zhu, J. Lou, Y. Liu, Y. Li, and Z. Wang, "Event-triggered control for the stabilization of probabilistic Boolean control networks," *Complexity*, vol. 2018, Sep. 2018, Art. no. 9259348.
- [7] B. Chen, J. Cao, G. Lu, and L. Rutkowski, "Stabilization of Markovian jump Boolean control networks via sampled-data control," *IEEE Trans. Cybern.*, to be published, doi: [10.1109/TCYB.2021.3069750](https://doi.org/10.1109/TCYB.2021.3069750).
- [8] B. Chen, J. Cao, J. Zhong, and L. Xiong, "Asymptotic stability of probabilistic logical networks with random impulsive effects," *Inf. Sci.*, vol. 579, pp. 667–684, 2021.
- [9] M. Meng, L. Liu, and G. Feng, "Stability and l_1 gain analysis of Boolean networks with Markovian jump parameters," *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 4222–4228, Aug. 2017.
- [10] X. Ding, H. Li, X. Li, and W. Sun, "Stability analysis of Boolean networks with stochastic function perturbations," *IEEE Access*, vol. 7, pp. 1323–1329, 2019.
- [11] H. Li, X. Xu, and X. Ding, "Finite-time stability analysis of stochastic switched Boolean networks with impulsive effect," *Appl. Math. Comput.*, vol. 347, pp. 557–565, 2019.
- [12] Y. Guo, R. Zhou, Y. Wu, W. Gui, and C. Yang, "Stability and set stability in distribution of probabilistic Boolean networks," *IEEE Trans. Autom. Control*, vol. 64, no. 2, pp. 736–742, Feb. 2019.
- [13] S. Zhu, J. Lu, Y. Lou, and Y. Liu, "Induced-equations-based stability analysis and stabilization of Markovian jump Boolean networks," *IEEE Trans. Autom. Control*, vol. 66, no. 10, pp. 4820–4827, Oct. 2021.
- [14] M. Meng and G. Xiao, "Output feedback control of Markovian switching Boolean control networks," in *Proc. 15th IEEE Int. Conf. Control Autom.*, 2019, pp. 830–835.
- [15] M. Meng, G. Xiao, C. Zhai, and G. Li, "Controllability of Markovian jump Boolean control networks," *Automatica*, vol. 106, pp. 70–76, 2019.
- [16] B. Chen, X. Yang, Y. Liu, and J. Qiu, "Controllability and stabilization of Boolean control networks by the auxiliary function of flipping," *Int. J. Robust Nonlinear Control*, vol. 30, no. 14, pp. 5529–5541, 2020.

- [17] H. Li, L. Xie, and Y. Wang, "Output regulation of Boolean control networks," *IEEE Trans. Autom. Control*, vol. 62, no. 6, pp. 2993–2998, Jun. 2017.
- [18] B. Chen, J. Cao, Y. Luo, and L. Rutkowski, "Asymptotic output tracking of probabilistic Boolean control networks," *IEEE Trans. Circuits Syst. I., Reg. Papers*, vol. 67, no. 8, pp. 2780–2790, Aug. 2020.
- [19] B. Li, J. Lu, Y. Liu, and Z.-G. Wu, "The outputs robustness of Boolean control networks via pinning control," *IEEE Trans. Control Netw. Syst.*, vol. 7, no. 1, pp. 201–209, Mar. 2020.
- [20] Y. Li, H. Li, and W. Sun, "Event-triggered control for robust set stabilization of logical control networks," *Automatica*, vol. 95, pp. 556–560, 2018.
- [21] T. Tian, K. Burrage, P. M. Burrage, and M. Carletti, "Stochastic delay differential equations for genetic regulatory networks," *J. Comput. Appl. Math.*, vol. 205, no. 2, pp. 696–707, 2007.
- [22] S. Kim *et al.*, "Can Markov chain models mimic biological regulation?," *J Biol. Syst.*, vol. 10, no. 4, pp. 337–357, 2002.
- [23] I. Shmulevich, E. R. Dougherty, and W. Zhang, "From Boolean to probabilistic Boolean networks as models of genetic regulatory networks," *Proc. IEEE*, vol. 90, no. 11, pp. 1778–1792, Nov. 2002.
- [24] P. Balasubramaniam, R. Rakkiyappan, and R. Krishnasamy, "Stochastic stability of Markovian jumping uncertain stochastic genetic regulatory networks with interval time-varying delays," *Math. Biosci.*, vol. 226, no. 2, pp. 97–108, 2010.
- [25] W. Zhang, J.-A. Fang, and Y. Tang, "Stochastic stability of Markovian jumping genetic regulatory networks with mixed time delays," *Appl. Math. Comput.*, vol. 217, no. 17, pp. 7210–7225, 2011.
- [26] Q. Li, B. Shen, Y. Liu, and F. E. Alsaadi, "Event-triggered H_∞ state estimation for discrete-time stochastic genetic regulatory networks with Markovian jumping parameters and time-varying delays," *Neurocomputing*, vol. 174, pp. 912–920, 2016.
- [27] D. Yue, Z.-H. Guan, T. Li, R.-Q. Liao, F. Liu, and Q. Lai, "Event-based cluster synchronization of coupled genetic regulatory networks," *Physica A. Statist. Mech. Appl.*, vol. 482, pp. 649–665, 2017.
- [28] W. P. M. H. Heemels, K. H. Johansson, and P. Tabuada, "An introduction to event-triggered and self-triggered control," in *Proc. 51st IEEE Annu. Conf. Decis. Control*, 2012, pp. 3270–3285.
- [29] B. Jiang, J. Lu, Y. Liu, and J. Cao, "Periodic event-triggered adaptive control for attitude stabilization under input saturation," *IEEE Trans. Circuits Syst. I., Reg. Papers*, vol. 67, no. 1, pp. 249–258, Jan. 2020.
- [30] P. Guo, H. Zhang, F. E. Alsaadi, and T. Hayat, "Semi-tensor product method to a class of event-triggered control for finite evolutionary networked games," *IET Control Theory Appl.*, vol. 11, no. 13, pp. 2140–2145, 2017.
- [31] Z. Wu, Q.-S. Jia, and X. Guan, "Optimal control of multiroom HVAC system: An event-based approach," *IEEE Trans. Control Syst. Technol.*, vol. 24, no. 2, pp. 662–669, Mar. 2016.
- [32] Q.-S. Jia, J. Wu, Z. Wu, and X. Guan, "Event-based HVAC control—a complexity-based approach," *IEEE Trans. Autom. Sci. Eng.*, vol. 15, no. 4, pp. 1909–1919, Oct. 2018.
- [33] Q.-S. Jia, J.-X. Tang, and Z. Lang, "Event-based optimization with random packet dropping," *Sci. China Inf. Sci.*, vol. 63, Nov. 2020, Art. no. 212202.
- [34] J. Edmonds, "Optimum branchings," *J. Res. Nat. Bur. Standards B*, vol. 71, no. 4, pp. 233–240, 1967.
- [35] C. G. Khatri and C. R. Rao, "Solutions to some functional equations and their applications to characterization of probability distributions," *Sankhyā Indian J. Statist., A*, vol. 30, no. 2, pp. 167–180, 1968.
- [36] S. M. Ross, *Stochastic Processes*. New York, NY, USA: Wiley, 1996.
- [37] K. Kobayashi and K. Hiraishi, "An integer programming approach to optimal control problems in context-sensitive probabilistic Boolean networks," *Automatica*, vol. 47, no. 6, pp. 1260–1264, 2011.
- [38] R. Tarjan, "Depth-first search and linear graph algorithms," *SIAM J. Comp.*, vol. 1, no. 2, pp. 146–160, 1972.